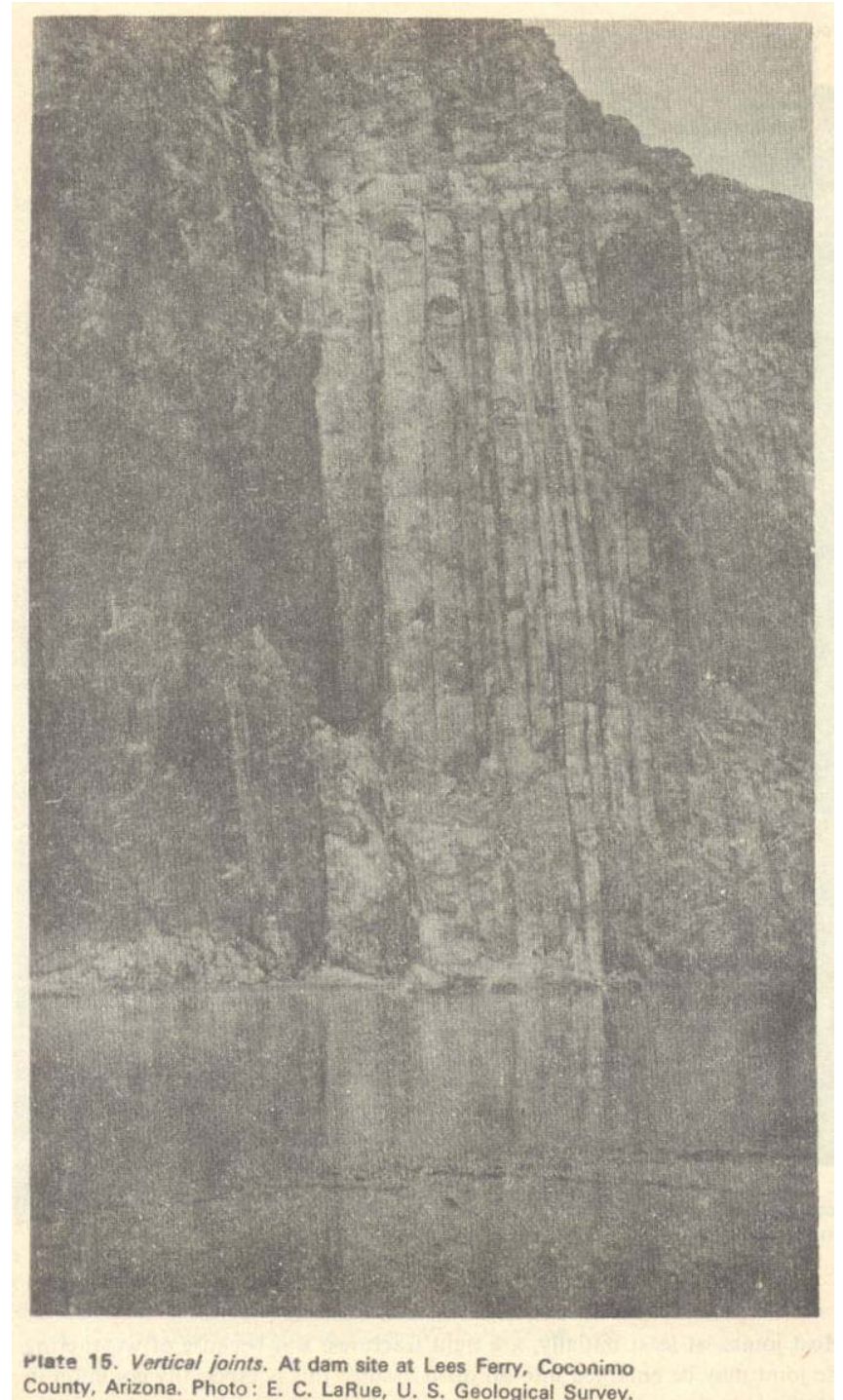


Joints

- Fracture that form by tensile loading
- **joints are smooth fractures along which rock breaks.**
- The length of such fractures (feet, tens of feet or 100s of feet).
- Most joints are planes, some curved also.
- No visible movement parallel to surface of joints. If movement noticed: they are faults.
- There may be very slight movement in some cases also.



- Slight movement at right angle to joint produces open fissures.
- Most joints are smooth, but some display **plumose markings**. These are ridges and depressions (~ mm) on a joint surface.

**Plumose
markings** ←

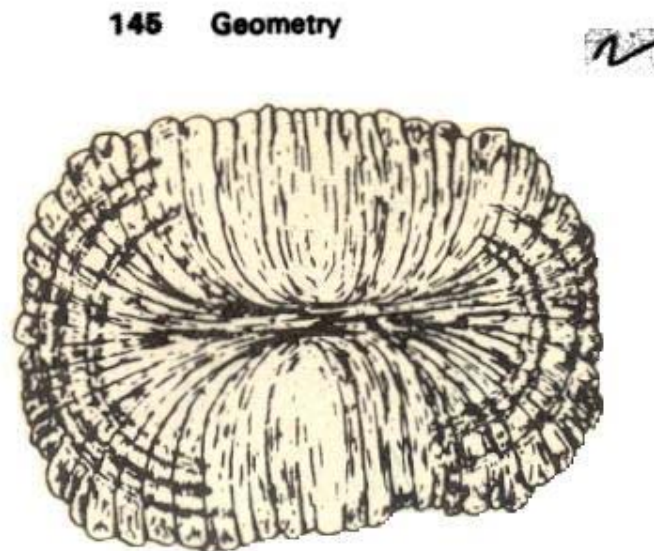


Fig. 7-1. Plumose and conchoidal structure on a joint several feet across. Comb Ridge–Navajo Mountain Area, Arizona and Utah. (After R. A. Hodgson.²) Permission American Association Petroleum Geologists.

Attitude of joints:

Strike: Direction of horizontal line on the surface of joints

Dip: angle between a hor.pl. and the joint measured in a vertical plane $\perp$ to strike of joint.

ABCD: Vertical dip, Striking EW

BDEF: Vertical dip, Striking NS

GHIJ: NS striking, Dipping 50°E

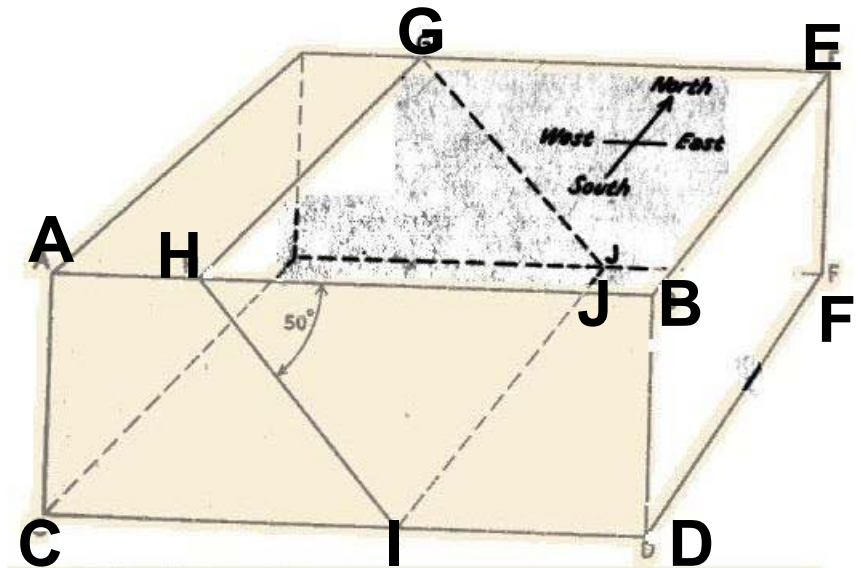


Fig. 7-2. Attitude of joints. Plane *ABCD* represents a vertical joint that strikes east-west; plane *BDEF* represents a vertical joint that strikes north-south; plane *GHIJ* represents a joint that strikes north-south and dips 50° east.

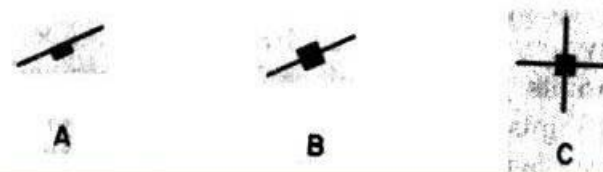


Fig. 7-3. Map symbols for joints. (A) Strike and dip of inclined joint. (B) Strike of vertical joint. (C) Horizontal joint.

Map symbols

Classification of joints:

2 schemes

- (i) Geometrical classification**
- (ii) Genetic classification**

Geometrical classification of joints:

- Strictly descriptive**
- Easy to apply**
- Doesn't indicate origin of joints**
- In this joints are classified on the basis of their attitude relative to bedding or similar structure in that rocks they cut.**

Strike joints:

are those that strike parallel to bedding of a Sedimentary rock /schistosity in a Schist.

eg. BDEF, MNO

Dip joints:

are those that strike parallel to dip direction of a bedding of a Sedimentary rock /schistosity or gneissose structure in a metamorphic rock.

eg. ABCD, GAI

Diagonal joints: are those that strike in direction Between strike and dip of associated rocks.

eg. PQR,STU

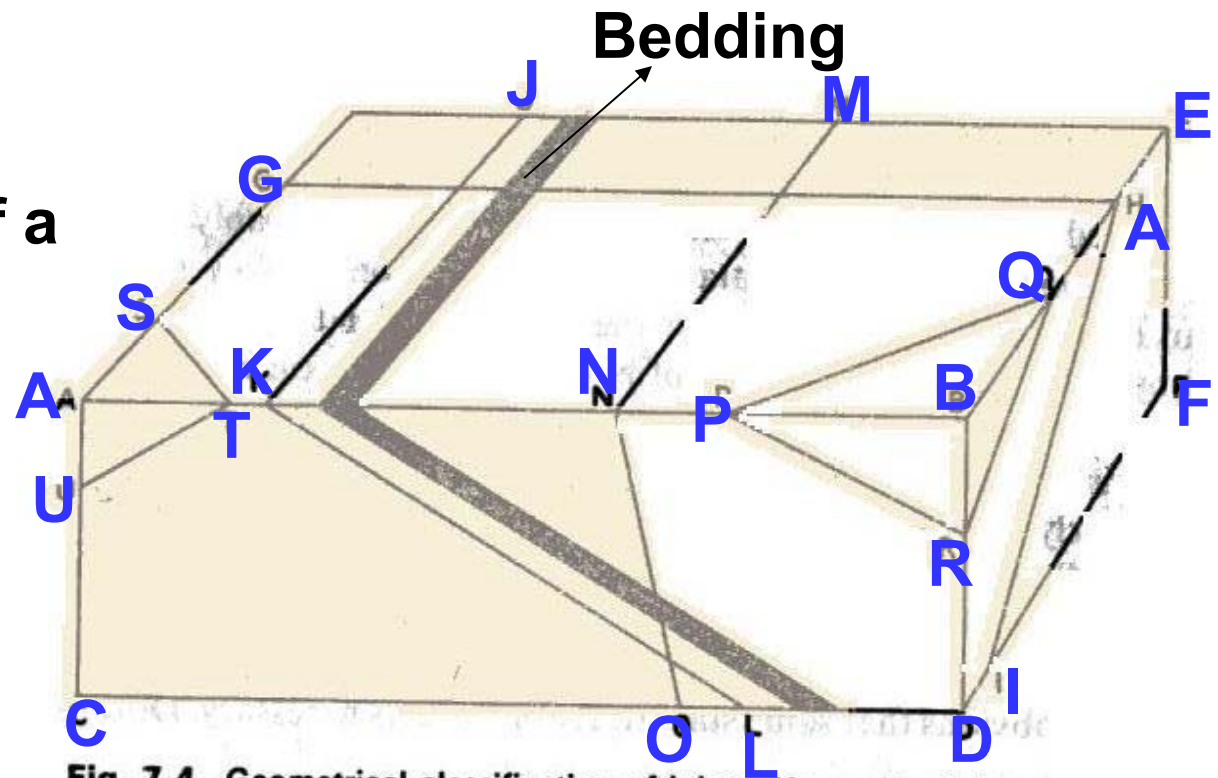


Fig. 7-4. Geometrical classification of joints. Heavy black layer is a bed. ABCD and GHI are dip joints; BDEF and MNO are strike joints. JKL is a bedding joint. PQR and STU are diagonal joints.

joint set : Group of more or less parallel joints

joint system : Consist of two or more joint sets or any group of joints with a characteristic pattern.

joint classification according to their strike:

North-south set, North-east set, east-west set etc. In some Region there may be one set vertical, a second set may be Dip 40°E and a third set may dip 60°W .

Block with three sets of joints ←

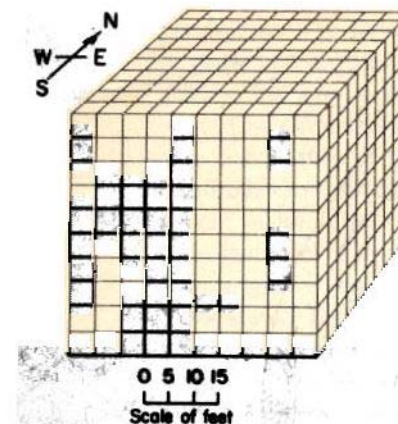


Fig. 7-7. Block with three sets of joints. Demonstrates importance of attitude of surface on which sets are seen.

Genetic classification of joints:

(i) Shear Fractures (ii) Tension fractures

Causes: (i) Tectonic stress causing fracturing coeval with tectonism

(ii) Residual stresses due to events that happened long before the fracturing

(iii) contraction due to shrinkage because of cooling or desiccation

(iv) surficial movements such as downhill movement of rocks or glaciers

joint vis-à-vis folds: Many joints are systematically disposed about folds and it has been assumed that they are result of same compressive forces as those resulted folds

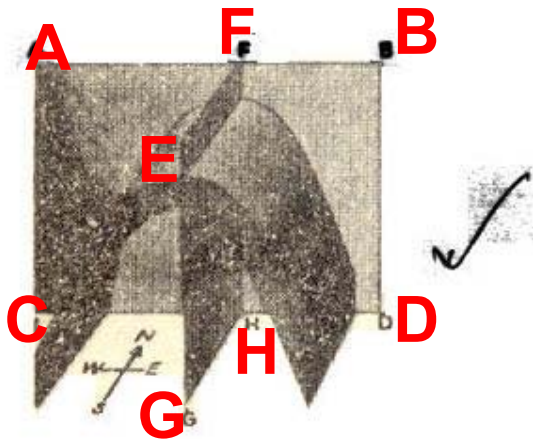


Fig. 7-25. Fold with vertical dip joint and vertical strike joint. *ABCD*, vertical dip joint. *EFGH*, vertical strike joint.

- joints perpendicular to axes of folds are common in orogenic belts (*ABCD*). These may be **extension joints** resulting from slight elongation parallel to axis of fold.

- joints parallel to those of axial planes of folds may be **release joints (EFGH)** developing from $\perp$ r to axis of compression when the load is released.

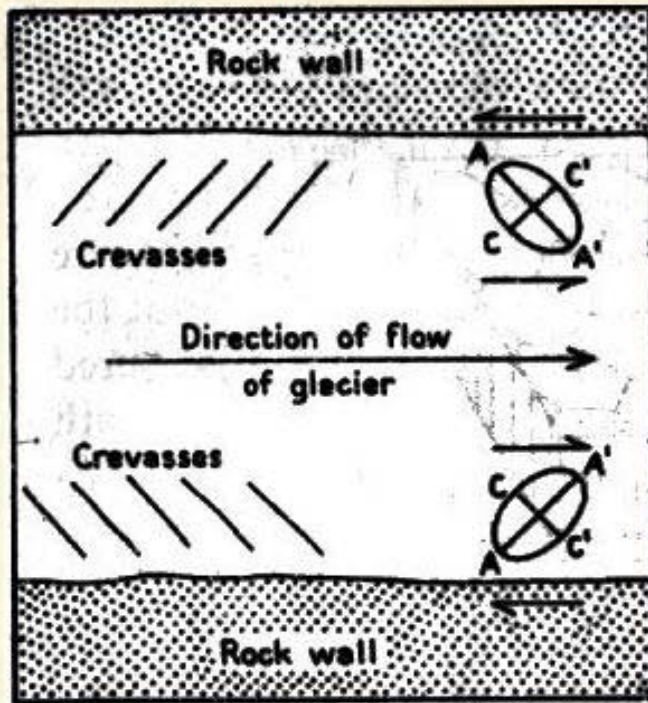
joints due to residual Stress:

- observed in massive granitic rocks. Such rocks consolidate at depths several miles in the crust under high lithostatic pressure. When are exposed to surface due to erosion they tend to expand and joints develop.
- (i) **Sheeting**: A form of rupture similar to jointing exposed in artificial opening such as quarries. The sheeting surfaces are somewhat curved and are normally parallel to surface of the earth.
- (ii) **Exfoliation domes**:

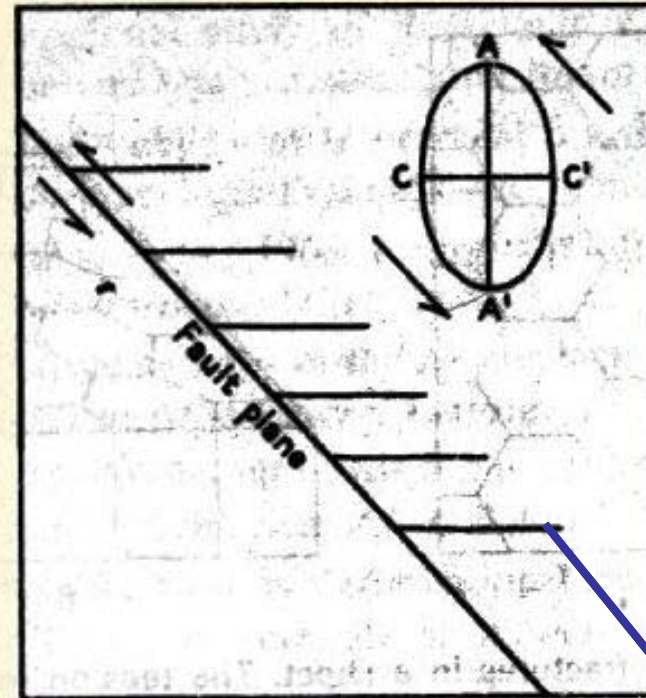
These form in granitic areas.

These are mountains composed of granite that is essentially un-jointed except for fractures that are parallel to surface of dome, dipping gently on the top but steeply on cliffs.

Tension fractures:



A



B

Fig. 7-27. Tension fractures. (A) Crevasses along side of glacier. Couples caused by friction are shown by smaller arrows. Ellipses represent orientation of strain ellipsoid. (B) Cross section to show feather joints, which are represented by horizontal lines to right of fault plane. Arrows near fault show relative movement along it. Orientation of ellipse resulting from this movement shown in upper right-hand corner.

Feather joints

Columnar jointing: In basaltic flows, sills and dykes.
These are contraction phenomena.

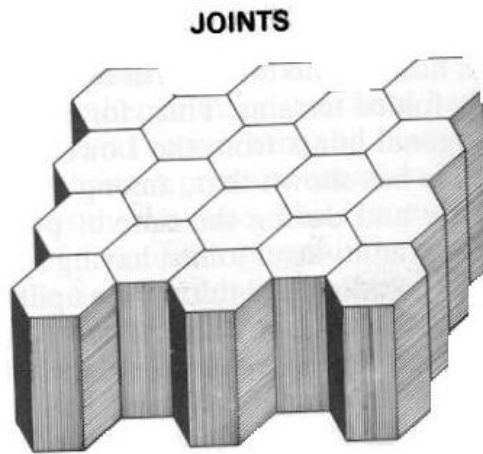
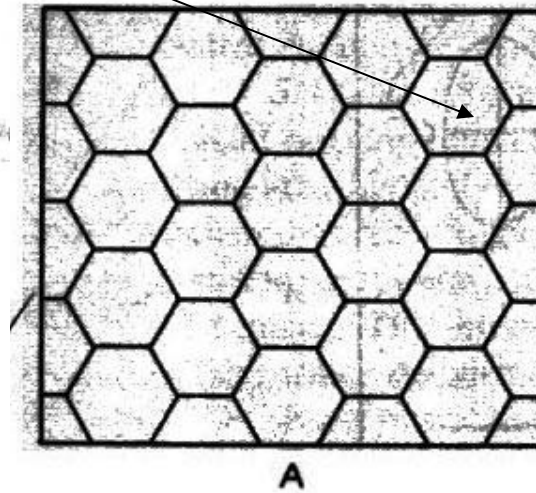
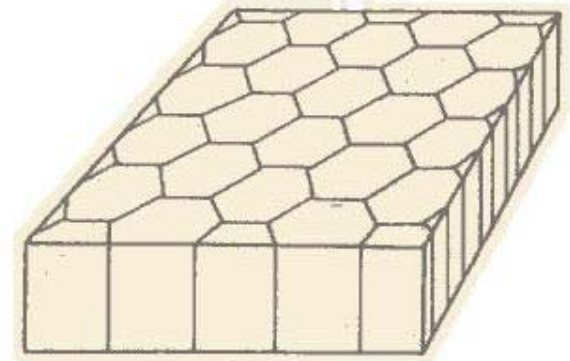


FIG. 19.13. Columnar joints.



A



B

Fig. 7-28. Hexagonal fractures in a sheet. The tension results from loss of volume either because of cooling of lava or desiccation of mud. (A) Upper surface of sheet. (B) Block diagram of sheet.

Columnar jointing:

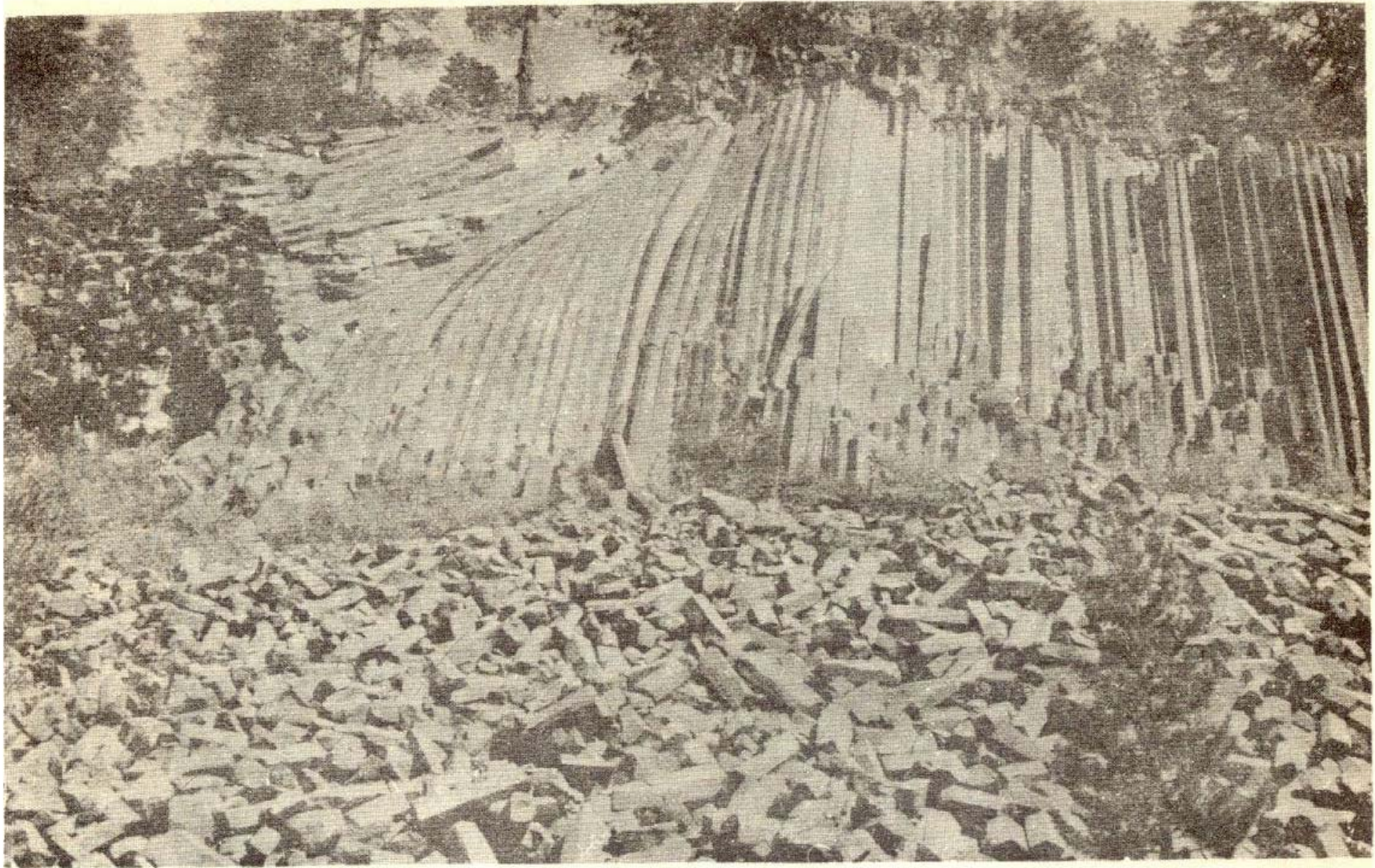


Plate 17B. *A columnar jointing.* Basalt flow. Devil Postpile National Monument. Madera County, California. Photo: H. R. Cornwall, U. S. Geological Survey.

Primary joints: (can be classed under genetic type)

- Develop during last stage of emplacement of an igneous intrusion when its outer part has been solidified but partially liquid inner part is still moving.
- Columnar joints in basalts are primary joints.

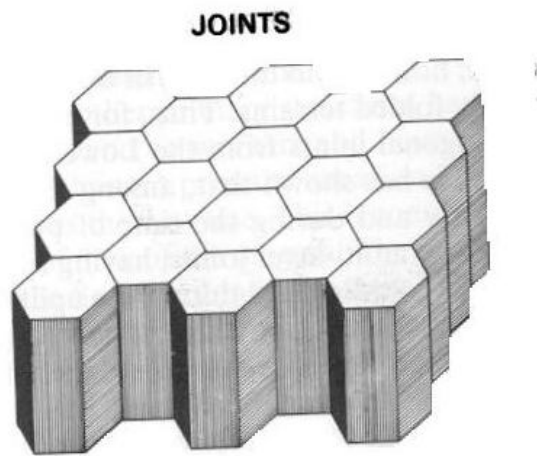


FIG. 19.13. Columnar joints.

Columnar joints:

- Most common in basaltic flows, sills and dykes.
- These form by shrinkage of the lava flow during its cooling.
- In a horizontal sheet of lava flow the joints appear as vertical fractures approximately at an angle of 120° with one another and dissect the into vertical columns with hexagonal cross sections.
- Such pattern also similar to that of mud cracks which form by shrinkage of mud during drying.

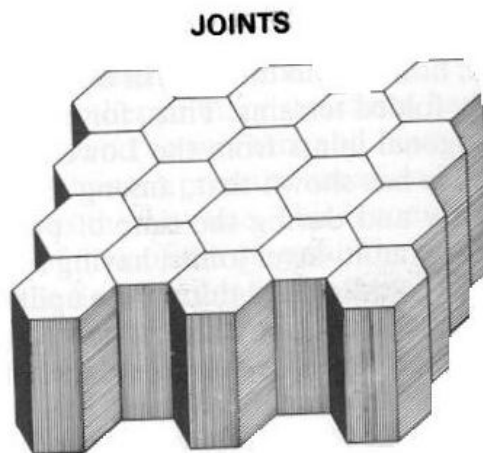
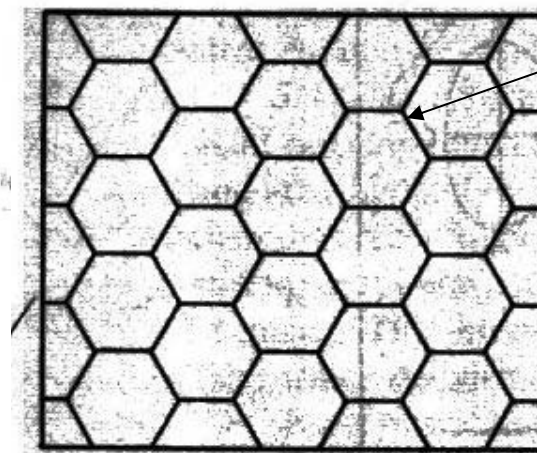
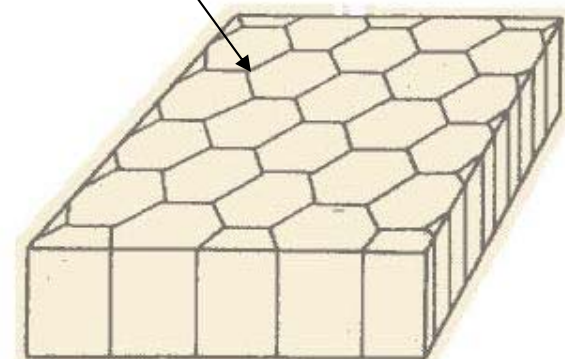


FIG. 19.13. Columnar joints.



A



B

Fig. 7-28. Hexagonal fractures in a sheet. The tension results from loss of volume either because of cooling of lava or desiccation of mud. (A) Upper surface of sheet. (B) Block diagram of sheet.

Plumose structures:

- Are one type of surface markings on joint surfaces.
- These are ridges and depressions (~ mm) on a joint surface.
- Plumose structures **record fracture propagation direction** in rocks.
- These structures are a **feather-like pattern of ridges and grooves** on the fracture surface.
- They form only under extensional forces and occur when a rapid, near-explosive snapping apart of the rock is encountered.



Plumose fracture in sandstone



Plumose fracture in sandstone

- forms when rock breaks.
- The fracture starts at one point and then grows, propagating thorough the rock and leaving behind a telling signature of its growth.



(In this case, the fracture (**i.e. joint**) started at point A and propagated through the rock to point B (central 'shaft'), expanding laterally (feathery 'plumes') at the same time).

Different parts of plumose structure

Origin: is the point of initiation of fracture

Mist: consists of tiny irregularities on the surface of this fracture and are arranged in a pattern which look like lateral veins of leaf.

Plume axis: is the axis from which individual bars propagate which also indicates the direction of propagation of joints

Twist hackle: consists of steps at the edge of fracture plane. The steps represent the zone in which the fracture has a split into a set of smaller *en echelon* fractures. The individual members of these individual fractures are not generally parallel to the main joint surface.

